Worst-Case Secrecy Rate Maximization for Fluid Antenna-aided Near-Field ISAC System

Zijian Lu*, Hengyi Liu†, Yiping Zuo‡§, Mingkai Chen*, Weibei Fan‡¶‖, Ning Gao§
* College of Telecommunications and Information Engineering, Nanjing University of Posts and Telecommunications, China
† College of Electronic and Optical Engineering, Nanjing University of Posts and Telecommunications, China
‡ College of Computer, Nanjing University of Posts and Telecommunications, Nanjing 210023, China
§ National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China
¶ State Key Laboratory of Internet Architecture, Tsinghua University, Beijing 100084, China
‖ State Key Lab of Processors, Institute of Computing Technology, CAS, China
Emails: 18818732360@163.com, lhy19121035781@outlook.com, zuoyiping@njupt.edu.cn
mkchen@njupt.edu.cn, wbfan@njupt.edu.cn, ninggao@seu.edu.cn

***Abstract*—Near-field secure integrated sensing and communication (ISAC) is particularly challenging under eavesdropper-location uncertainty and sensing-accuracy requirements, because secrecy performance then depends on both signaling and array geometry. Existing secure ISAC designs still mainly rely on fixed arrays and optimize only waveform, beamforming, or artificial noise (AN), which becomes insufficient when eavesdropper locations are uncertain and sensing accuracy must be preserved simultaneously. We investigate a monostatic near-field secure ISAC system with fluid antennas (FAs), where a base station serves users, senses targets, and suppresses eavesdroppers with imperfectly known locations. Under a Cramér–Rao-bound (CRB) constraint, we formulate a geometry-aware secrecy design that jointly optimizes the transmit and receive FA positions, the communication beamformer, and the AN covariance and power. To handle the resulting nonconvex coupling among geometry, secrecy leakage, sensing accuracy, and AN design, we develop a mixed particle-swarm-optimization, sine-cosine-search, and alternating-optimization (PSOSC)-based algorithm. Numerical results verify stable convergence under uncertainty and show that FA reconfiguration enlarges the secrecy sensing operating region over fixed layouts, while directional AN and larger transmit arrays provide additional secrecy gains.**

***Index Terms*—Fluid antenna, ISAC, near-field, secrecy rate, AN.**

I. Introduction

Integrated sensing and communication (ISAC) becomes particularly challenging when confidential transmission, target sensing, and suspicious listeners coexist in the same deployment [1]–[3]. In near-field secure ISAC, performance is jointly shaped by signaling and array geometry, because geometry directly affects both information leakage and sensing accuracy. Once geometry enters the secrecy mechanism, secure ISAC must be treated as a coupled secrecy-sensing design problem rather than a conventional fixed-array beamforming task.

Most existing secure ISAC studies still optimize waveform design, interference management, beamforming, or artificial noise (AN) over fixed arrays [4]–[8]. These designs are important, but they become insufficient in the near field, where geometry itself affects channel responses and leakage patterns. Meanwhile, Cramér-Rao-bound (CRB) driven sensing formulations show that sensing accuracy is also geometry dependent [9]. Therefore, in near-field secure ISAC, geometry cannot be designed for secrecy alone. It must be optimized under sensing-accuracy requirements as well.

The problem becomes more challenging when eavesdropper locations are imperfectly known, since a geometry that is favorable for a nominal position may become vulnerable under location mismatch. Fluid antenna (FA) technology provides a new spatial degree of freedom by allowing each antenna element to move inside a local region [10], [11], and related movable array studies have shown its potential in communication and physical layer security [12], [13]. In parallel, existing dual-functional artificial-noise (DFAN) results indicate that directional AN shaping is valuable for secrecy enhancement in integrated sensing settings [14]. However, these mechanisms have not been jointly exploited with geometry reconfiguration, uncertainty handling, and sensing-constrained design in the near field.

More importantly, these ingredients cannot be handled in isolation in near-field secure ISAC. Once the array geometry is reconfigurable, the propagation structure, secrecy leakage, sensing accuracy, and AN effectiveness all become coupled through the same spatial configuration. A geometry that improves the legitimate link may also reshape the eavesdropping channel and the sensing echo, while a stronger AN design may alter the power left for communication and sensing. Therefore, the problem is not merely one of adding geometry optimization to secure beamforming, but of jointly designing

Corresponding author: Yiping Zuo (email: zuoyiping@njupt.edu.cn).

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geometry and signaling under secrecy sensing coupling.

Motivated by these observations, we study a monostatic near-field secure ISAC system with movable transmit and receive FAs under bounded eavesdropper-location uncertainty. We maximize a sampled worst-case average secrecy metric under CRB, power, region, and spacing constraints by jointly optimizing the FA positions, communication beamforming, and directional AN design. The resulting problem is highly nonconvex because geometry-channel coupling, secrecy-sensing coupling, and beamforming/AN/FA-position coupling coexist in the same formulation. To address it, we develop an outer-inner particle-swarm-optimization, sine-cosine-search, and alternating-optimization (PSOSC) framework, in which particle swarm optimization (PSO) searches the FA geometry and an inner sine-cosine-search (SC) and alternating-optimization (AO) stage refines the beamformer and AN design, and the main contributions are summarized as follows:

- We establish a near-field secure ISAC model under bounded eavesdropper-location uncertainty with movable transmit and receive FAs, where sampled secrecy leakage and CRB-constrained sensing accuracy are jointly embedded into a geometry-aware robust design.
- We develop a hybrid PSOSC algorithm that jointly optimizes FA positions, communication beamforming, and directional AN, with outer PSO geometry search and inner AO-based signal refinement under each candidate layout.
- Numerical results show that the proposed design remains convergent under uncertainty, that FA reconfiguration enlarges the secrecy sensing operating region over fixed layouts, and that directional AN with larger transmit arrays further amplifies the geometry enabled gain.

II. System Model

We consider a monostatic base station (BS) serving K legitimate users, sensing Q targets, and suppressing L eavesdroppers, where $\mathcal{K} = \{1,\dots,K\}$, $\mathcal{Q} = \{1,\dots,Q\}$, and $\mathcal{L} = \{1,\dots,L\}$. The BS employs M transmit FAs and N receive FAs, both movable inside square regions $\mathcal{D}_t = \mathcal{D}_r = [-A/2, A/2]^2$, while the minimum spacing between any two FAs on the same side is constrained by d_0. Denote the m-th transmit-FA position and the nth receive-FA position at slot t by $\mathbf{d}_{t,m}(t) = [x_{t,m}(t), y_{t,m}(t)]^T$ and $\mathbf{d}_{r,n}(t) = [x_{r,n}(t), y_{r,n}(t)]^T$, respectively, and stack them into $\mathbf{D}_t(t)$ and $\mathbf{D}_r(t)$. The user, eavesdropper, and target locations are denoted by $\mathbf{p}_k^{(U)}(t)$, $\mathbf{p}_\ell^{(E)}(t)$, and $\mathbf{p}_q^{(T)}(t)$. All node positions are assumed quasi-static within one slot.

A. Communication and Secrecy Model

As shown in Fig. 1, the BS sends confidential information while injecting dual-functional AN for eavesdropper suppression and target illumination.

At slot t, the transmit signal is

$$\mathbf{x}(t) = \sum_{k\in\mathcal{K}} \mathbf{w}_k(t)s_k(t) + \sqrt{p_a(t)}\,\mathbf{z}_a(t), \tag{1}$$

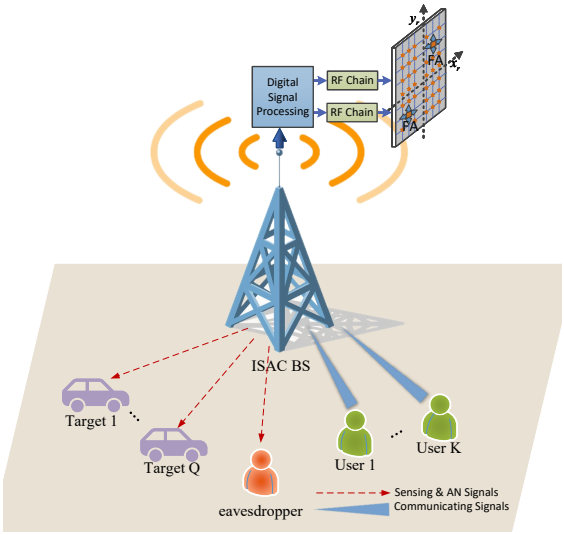


Fig. 1. FA-assisted near-field secure ISAC architecture.

where $s_k(t) \sim \mathcal{CN}(0,1)$ is the symbol intended for user k, $\mathbf{W}_c(t) = [\mathbf{w}_1(t),\dots,\mathbf{w}_K(t)]$ collects the communication beamformers, $p_a(t)$ is the AN power, and $\mathbf{z}_a(t) \sim \mathcal{CN}(\mathbf{0}, \mathbf{D}_a(t))$ with $\mathrm{Tr}(\mathbf{D}_a(t)) = 1$. The received signal at user k is

$$y_k(t) = \mathbf{h}_k^H(t)\mathbf{x}(t) + n_k(t), \tag{2}$$

where $n_k(t) \sim \mathcal{CN}(0,\sigma_k^2)$ is the additive noise at user k.

Under the near-field non-uniform spherical-wave model, the channel from the transmit FA array to user k is

$$\begin{aligned} \mathbf{h}_k(t) &= \frac{\sqrt{\alpha}}{r_k(t)}\mathbf{a}_k(t), \\ [\mathbf{a}_k(t)]_m &= \frac{r_k(t)}{r_{k,m}^{(\mathrm{tx})}(t)} \exp\!\left(-j\frac{2\pi}{\lambda}\big(r_{k,m}^{(\mathrm{tx})}(t) - r_k(t)\big)\right). \end{aligned} \tag{3}$$

where $\alpha = (\lambda/4\pi)^2$, $r_k(t) = \|\mathbf{p}_k^{(U)}(t)\|$, and $r_{k,m}^{(\mathrm{tx})}(t) = \|[\mathbf{p}_k^{(U)}(t)]^T - [\mathbf{d}_{t,m}^T(t), 0]^T\|$. Accordingly, the m-th channel coefficient is given by

$$[\mathbf{h}_k(t)]_m = \frac{\sqrt{\alpha}}{r_{k,m}^{(\mathrm{tx})}(t)} \exp\!\left(-j\frac{2\pi}{\lambda}\big(r_{k,m}^{(\mathrm{tx})}(t) - r_k(t)\big)\right). \tag{4}$$

Because the DFAN waveform is pre-shared, the legitimate user can cancel it before decoding [15]. Hence the user signal-to-interference-plus-noise ratio (SINR) is

$$\gamma_k(t) = \frac{|\mathbf{h}_k^H(t)\mathbf{w}_k(t)|^2}{\sum_{i\neq k}|\mathbf{h}_k^H(t)\mathbf{w}_i(t)|^2 + \sigma_k^2}, \tag{5}$$

For eavesdropper ℓ, we assume

$$\mathbf{p}_\ell^{(E)}(t) = \hat{\mathbf{p}}_\ell^{(E)}(t) + \Delta_\ell(t), \quad \|\Delta_\ell(t)\| \le \rho_e, \tag{6}$$

and sample the corresponding uncertainty disk by S_e support points $\{\mathbf{p}_{\ell,s}^{(E)}(t)\}_{s=1}^{S_e}$. At support point s, the received signal is

$$y_{\ell,s}^{(E)}(t) = \mathbf{g}_{\ell,s}^H(t)\mathbf{x}(t) + n_{\ell,s}^{(E)}(t), \tag{7}$$

where $n_{\ell,s}^{(E)}(t) \sim \mathcal{CN}(0, \sigma_e^2)$ is the additive noise at support point s. The near-field channel from the BS to that support point satisfies

$$[\mathbf{g}_{\ell,s}(t)]_m = \frac{\sqrt{\alpha}}{r_{\ell,s,m}^{(\mathrm{tx})}(t)} \exp\!\left(-j\frac{2\pi}{\lambda}\big(r_{\ell,s,m}^{(\mathrm{tx})}(t) - r_{\ell,s}(t)\big)\right), \tag{8}$$

where $r_{\ell,s}(t) = \|\mathbf{p}_{\ell,s}^{(E)}(t)\|$ and $r_{\ell,s,m}^{(\mathrm{tx})}(t) = \|[\mathbf{p}_{\ell,s}^{(E)}(t)]^T - [\mathbf{d}_{t,m}^T(t), 0]^T\|$. Therefore, the eavesdropping SINR for decoding user k is

$$\begin{aligned}\gamma_{\ell,k,s}^{(E)}(t) = |\mathbf{g}_{\ell,s}^H(t)\mathbf{w}_k(t)|^2 \Big/ \Big(\sum_{i\neq k} |\mathbf{g}_{\ell,s}^H(t)\mathbf{w}_i(t)|^2 \\ + p_a(t)\mathbf{g}_{\ell,s}^H(t)\mathbf{D}_a(t)\mathbf{g}_{\ell,s}(t) + \sigma_e^2\Big).\end{aligned} \tag{9}$$

The classical pairwise secrecy rate is then

$$\begin{aligned}R_{k,\ell}^{\mathrm{sec}}(t) = \Big[\log_2(1+\gamma_k(t)) \\ - \max_{s\in\{1,\ldots,S_e\}} \log_2(1+\gamma_{\ell,k,s}^{(E)}(t))\Big].\end{aligned} \tag{10}$$

To account for eavesdropper-location uncertainty, we evaluate the leakage term at the worst sampled support point for each user–eavesdropper pair (k, ℓ). The implemented solver then optimizes the sampled worst-case average secrecy metric, denoted by R_s^{worst},

$$\begin{aligned}R_s^{\mathrm{worst}} = \frac{1}{KL}\sum_{k\in\mathcal{K}}\sum_{\ell\in\mathcal{L}}\Big(\log_2(1+\gamma_k) \\ - \max_{s\in\{1,\ldots,S_e\}} \log_2(1+\gamma_{\ell,k,s}^{(E)})\Big),\end{aligned} \tag{11}$$

where the inner maximization captures the worst sampled leakage for each pair, while the outer averaging prevents the design from being dominated by a single extreme user–eavesdropper pair.

B. Sensing Model and CRB Metric

The receive FA array is also used for target sensing. In one slot, the echo observed at the BS can be written as

$$\mathbf{y}_R = \sum_{q=1}^{Q} \lambda_q \mathbf{a}_q(\mathbf{D}_r)\mathbf{b}_q^H(\mathbf{D}_t)\mathbf{x} + \mathbf{n}_R = \mathbf{A}\mathbf{\Lambda}\mathbf{B}^H\mathbf{x} + \mathbf{n}_R, \tag{12}$$

where λ_q is the round-trip complex gain of target q, $\mathbf{n}_R \sim \mathcal{CN}(\mathbf{0}, \sigma_R^2\mathbf{I})$ is the additive noise vector at the BS receiver, $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_Q] \in \mathbb{C}^{N\times Q}$ collects the receive steering vectors, and $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_Q] \in \mathbb{C}^{M\times Q}$ collects the transmit steering vectors.

Under the near-field spherical-wave model, the nth entry of $\mathbf{a}_q$ and the mth entry of $\mathbf{b}_q$ are

$$[\mathbf{a}_q]_n = \frac{r_q}{r_{q,n}^{(\mathrm{rx})}} \exp\!\left(-j\frac{2\pi}{\lambda}\big(r_{q,n}^{(\mathrm{rx})} - r_q\big)\right), \tag{13}$$

$$[\mathbf{b}_q]_m = \frac{r_q}{r_{q,m}^{(\mathrm{tx})}} \exp\!\left(-j\frac{2\pi}{\lambda}\big(r_{q,m}^{(\mathrm{tx})} - r_q\big)\right), \tag{14}$$

where r_q and θ_q denote the nominal distance and angle of target q, and $r_{q,n}^{(\mathrm{rx})}$ and $r_{q,m}^{(\mathrm{tx})}$ are the distances from target q to the nth receive FA and the mth transmit FA, respectively.

Let the transmit covariance be

$$\mathbf{R}_x = \mathbf{W}_c\mathbf{W}_c^H + \mathbf{R}_a, \tag{15}$$

where $\mathbf{W}_c = [\mathbf{w}_1, \ldots, \mathbf{w}_K]$ denotes the communication beamforming matrix and $\mathbf{R}_a = p_a\mathbf{D}_a$ is the AN covariance matrix. We define the unknown sensing parameter vector as

$$\boldsymbol{\xi} = \big[\boldsymbol{\theta}^T, \mathbf{r}^T, \Re\{\boldsymbol{\lambda}\}^T, \Im\{\boldsymbol{\lambda}\}^T\big]^T, \tag{16}$$

where $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_Q]^T$, $\mathbf{r} = [r_1, \ldots, r_Q]^T$, and $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_Q]^T$.

Following the standard deterministic CRB construction, the Fisher information matrix can be organized into the block form

$$\mathbf{J} = \begin{bmatrix}\mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{12}^T & \mathbf{J}_{22}\end{bmatrix}, \tag{17}$$

where the noiseless sensing operator is

$$\mathbf{S}(\boldsymbol{\xi}) = \mathbf{A}(\boldsymbol{\theta}, \mathbf{r})\mathbf{\Lambda}\mathbf{B}^H(\boldsymbol{\theta}, \mathbf{r}). \tag{18}$$

For the Gaussian observation model in (12), the normalized FIM used in (17) can be written entrywise as

$$[\mathbf{J}]_{i,j} = \Re\left\{\mathrm{Tr}\left(\frac{\partial\mathbf{S}(\boldsymbol{\xi})}{\partial\xi_i}\mathbf{R}_x\frac{\partial\mathbf{S}^H(\boldsymbol{\xi})}{\partial\xi_j}\right)\right\}, \tag{19}$$

where ξ_i denotes the ith entry of $\boldsymbol{\xi}$. To make the dependence on the target angle and range explicit, we further define

$$\dot{\mathbf{A}}_\theta = \left[\frac{\partial\mathbf{a}_1}{\partial\theta_1}, \ldots, \frac{\partial\mathbf{a}_Q}{\partial\theta_Q}\right], \qquad \dot{\mathbf{A}}_r = \left[\frac{\partial\mathbf{a}_1}{\partial r_1}, \ldots, \frac{\partial\mathbf{a}_Q}{\partial r_Q}\right], \tag{20}$$

$$\dot{\mathbf{B}}_\theta = \left[\frac{\partial\mathbf{b}_1}{\partial\theta_1}, \ldots, \frac{\partial\mathbf{b}_Q}{\partial\theta_Q}\right], \qquad \dot{\mathbf{B}}_r = \left[\frac{\partial\mathbf{b}_1}{\partial r_1}, \ldots, \frac{\partial\mathbf{b}_Q}{\partial r_Q}\right]. \tag{21}$$

Accordingly, the three blocks in (17) admit the partition

$$\begin{aligned}\mathbf{J}_{11} &= \begin{bmatrix}\mathbf{J}_{\theta\theta} & \mathbf{J}_{\theta r} \\ \mathbf{J}_{r\theta} & \mathbf{J}_{rr}\end{bmatrix}, \\ \mathbf{J}_{12} &= \begin{bmatrix}\mathbf{J}_{\theta\lambda_R} & \mathbf{J}_{\theta\lambda_I} \\ \mathbf{J}_{r\lambda_R} & \mathbf{J}_{r\lambda_I}\end{bmatrix}, \\ \mathbf{J}_{22} &= \begin{bmatrix}\mathbf{J}_{\lambda_R\lambda_R} & \mathbf{J}_{\lambda_R\lambda_I} \\ \mathbf{J}_{\lambda_I\lambda_R} & \mathbf{J}_{\lambda_I\lambda_I}\end{bmatrix}.\end{aligned} \tag{22}$$

where $\boldsymbol{\lambda}_R = \Re\{\boldsymbol{\lambda}\}$ and $\boldsymbol{\lambda}_I = \Im\{\boldsymbol{\lambda}\}$. Each subblock is obtained from (19) by taking the corresponding derivatives of $\mathbf{A}$, $\mathbf{B}$, and $\boldsymbol{\lambda}$. Here, $\mathbf{J}_{11}$ corresponds to the location parameters $(\boldsymbol{\theta}, \mathbf{r})$, and $\mathbf{J}_{22}$ corresponds to the real and imaginary parts of the target gains. Using the steering-vector derivatives $\dot{\mathbf{A}}_r$, $\dot{\mathbf{A}}_\theta$, $\dot{\mathbf{B}}_r$, and $\dot{\mathbf{B}}_\theta$, the location-only CRB is obtained through the Schur complement

$$\mathbf{CRB}_{\mathrm{loc}} = \frac{\sigma_R^2}{2}\left(\mathbf{J}_{11} - \mathbf{J}_{12}\mathbf{J}_{22}^{-1}\mathbf{J}_{12}^T\right)^{-1}. \tag{23}$$

The sensing constraint used in the solver is therefore

$$\mathrm{Tr}(\mathbf{CRB}_{\mathrm{loc}}) \le \Gamma. \tag{24}$$

III. Problem Formulation and PSOSC-Based Algorithm

Based on the communication and sensing models above, we now formulate the joint design problem under eavesdropper-location uncertainty and organize its variables and subproblems in a way that directly matches the subsequent solver development.

The resulting optimization problem is

$$\begin{aligned}\mathcal{P}1: \quad & \max_{\mathbf{W}_c,\mathbf{D}_t,\mathbf{D}_r,p_a,\mathbf{D}_a} \quad R_s^{\mathrm{worst}} \\ & \text{C1: } \mathrm{Tr}(\mathbf{CRB}_{\mathrm{loc}}) \le \Gamma \\ & \text{C2: } 0 \le p_a \le P_{\max} - \sum_{k\in\mathcal{K}} \|\mathbf{w}_k\|^2 \\ & \text{C3: } \mathbf{D}_a \succeq \mathbf{0},\ \mathrm{Tr}(\mathbf{D}_a) = 1 \\ & \text{C4: } \mathbf{d}_{t,m} \in \mathcal{D}_t,\ \mathbf{d}_{r,n} \in \mathcal{D}_r \\ & \text{C5: } \|\mathbf{d}_{t,m} - \mathbf{d}_{t,m'}\| \ge d_0,\ \forall m \ne m' \\ & \text{C6: } \|\mathbf{d}_{r,n} - \mathbf{d}_{r,n'}\| \ge d_0,\ \forall n \ne n'\end{aligned} \tag{25}$$

where $\mathcal{X} \triangleq \{\mathbf{W}_c, \mathbf{D}_a, p_a, \mathbf{D}_t, \mathbf{D}_r\}$ denotes the decision-variable collection. Here, $\mathbf{W}_c$ denotes the communication beamforming matrix, $\mathbf{D}_a$ and p_a denote the normalized AN covariance shape and the scalar AN power, and $\mathbf{D}_t$ and $\mathbf{D}_r$ collect the transmit and receive FA coordinates, respectively.

Problem $\mathcal{P}1$ in (25) is difficult for three main reasons. First, the sampled worst-case average secrecy objective is nonconcave and depends on near-field propagation, AN covariance shaping, and the CRB expression in a highly coupled manner. Second, the signal-design block $\{\mathbf{W}_c, \mathbf{D}_a, p_a\}$ and the geometry block $\{\mathbf{D}_t, \mathbf{D}_r\}$ are strongly coupled through both communication and sensing channels. Third, the joint transmit/receive FA reconfiguration creates a high-dimensional feasible region with deployment and spacing constraints. Accordingly, directly solving (25) is intractable by standard convex tools.

To solve problem $\mathcal{P}1$, we partition the variables into three coupled blocks and update them alternately. With the other variables fixed, the resulting subproblems are given as follows.

$$\mathcal{P}1.1: \max_{\mathbf{W}_c} \ R_s^{\mathrm{worst}} \quad \text{s.t. C1 and C2,} \tag{26a}$$

$$\mathcal{P}1.2: \max_{\mathbf{D}_a,p_a} \ R_s^{\mathrm{worst}} \quad \text{s.t. C1, C2, and C3,} \tag{26b}$$

$$\mathcal{P}1.3: \max_{\mathbf{D}_t,\mathbf{D}_r} \ R_s^{\mathrm{worst}} \quad \text{s.t. C1, C2, and C4–C6.} \tag{26c}$$

Here, $\mathcal{P}1.1$ updates $\mathbf{W}_c$, $\mathcal{P}1.2$ updates $\mathbf{D}_a$ and p_a, and $\mathcal{P}1.3$ updates the FA positions. Under each candidate geometry, an inner AO loop alternates between $\mathcal{P}1.1$ and $\mathcal{P}1.2$ until convergence or the maximum inner-iteration budget is reached.

Subproblem $\mathcal{P}1.1$ updates the beamformer by SC search under fixed FA layouts and AN power. Subproblem $\mathcal{P}1.2$ updates the AN covariance and power allocation. Since the legitimate users can cancel the pre-shared AN locally, we first stack all support-point eavesdropper channels into $\mathbf{G}_{\mathrm{all}}$ and form the leakage covariance

$$\mathbf{S}_{\mathrm{an}} = \mathbf{G}_{\mathrm{all}}\mathbf{G}_{\mathrm{all}}^H. \tag{27}$$

Let $\mathbf{S}_{\mathrm{an}} = \mathbf{V}\,\mathrm{diag}(\mathbf{d})\mathbf{V}^H$ be its eigendecomposition. The AN covariance shape is constructed as

$$\mathbf{D}_a = \frac{\mathbf{V}\,\mathrm{diag}(\mathbf{d}^{\kappa})\,\mathbf{V}^H}{\mathrm{Tr}(\mathbf{V}\,\mathrm{diag}(\mathbf{d}^{\kappa})\,\mathbf{V}^H)}, \tag{28}$$

where $\kappa \ge 1$ is a fixed focusing exponent. After obtaining $\mathbf{D}_a$, the residual AN power is optimized over $[0, P_{\max} - \sum_{k\in\mathcal{K}} \|\mathbf{w}_k\|^2]$ by one-dimensional search.

Subproblem $\mathcal{P}1.3$ performs PSO-based FA reconfiguration. The transmit and receive FA coordinates are stacked into one particle vector, and the outer PSO updates this vector under the deployment-region and spacing constraints. For each candidate particle, the solver runs the inner AO loop formed by $\mathcal{P}1.1$ and $\mathcal{P}1.2$ and uses the resulting objective value as the particle fitness.

Combining the outer PSO geometry search in $\mathcal{P}1.3$ with the inner SC-AO updates in $\mathcal{P}1.1$ and $\mathcal{P}1.2$ yields the complete PSOSC-based procedure summarized in Algorithm 1. If the outer PSO uses N_p particles and I_p iterations, the inner AO loop runs for at most I_{AO} iterations, and each SC update uses N_s agents and I_s iterations, then the dominant complexity is on the order of $\mathcal{O}(N_p I_p I_{\mathrm{AO}} N_s I_s M K L S_e)$.

Algorithm 1 Mixed PSOSC-based Alternating Iterative Algorithm for Worst-Case Secrecy-Rate Maximization

Input: $P_{\max}$, Γ, $\mathcal{D}_t$, $\mathcal{D}_r$, d_0, and S_e.
Output: $\mathbf{W}_c^\star$, $\mathbf{D}_a^\star$, $p_a^\star$, $\mathbf{D}_t^\star$, and $\mathbf{D}_r^\star$.
Initialization: Generate eavesdropper support points, initialize feasible FA layouts, and choose an initial beamformer and AN power.
for each outer PSO iteration **do**
  for each particle **do**
    Decode the FA positions and initialize the inner state.
    repeat
      Stage I: Solve $\mathcal{P}1.1$ to update $\mathbf{W}_c$ by SC search.
      Stage II: Solve $\mathcal{P}1.2$ to update $\mathbf{D}_a$ and p_a.
    until the secrecy improvement is below a tolerance or the maximum number of inner iterations is reached
    Evaluate the particle fitness.
  end for
  Stage III: Solve $\mathcal{P}1.3$ to update particle bests, the global best, and the candidate FA layouts by PSO.
end for
Return the best beamformer, AN design, and FA layout.

IV. Numerical Results

Unless otherwise specified, all results are generated under the setting in Table I. Figures 2–4 together show that FA

TABLE I
SIMULATION PARAMETERS USED IN THE REPORTED RESULTS.

Parameter	Value
Legitimate users K	3
Eavesdroppers L	2
Targets Q	2
Transmit FAs M	4
Receive FAs N	4
λ	0.1 m
$\mathcal{D}_t, \mathcal{D}_r$	$[-0.5, 0.5] \times [-0.5, 0.5]$ m^2
d_0	0.5λ
$\lvert\lambda_q\rvert$	1
$P_{\max}$	1 W
ρ_e	2λ
S_e	8
Noise PSD	-174 dBm/Hz
Γ	10^{-12}

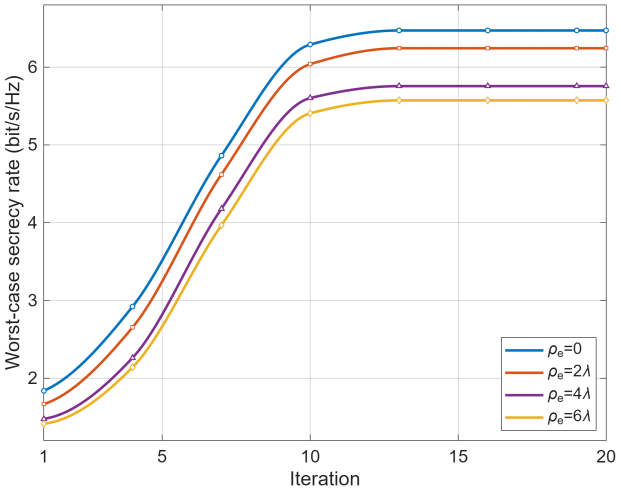


Fig. 2. Worst-case secrecy-rate convergence under different uncertainty radii.

reconfiguration turns geometry into an effective robustness resource for near-field secure ISAC: Fig. 2 verifies stable convergence under uncertainty, Fig. 3 shows that geometry reconfiguration enlarges the secrecy-sensing operating region, and Fig. 4 further demonstrates that directional AN and larger transmit arrays strengthen this gain.

A. Convergence Behavior

Fig. 2 shows that the proposed geometry aware robust design converges stably across the outer PSO iterations, where each point is obtained after the corresponding particle evaluations have completed their inner AO updates. The secrecy rate rises rapidly in the early iterations and then gradually levels off, indicating that the coupled geometry and signal-design procedure remains tractable even under location uncertainty. The final convergence level decreases as the eavesdropper uncertainty radius grows, because a larger uncertainty disk enlarges the vulnerable leakage region and forces the design to spend more degrees of freedom on robustness rather than pure legitimate-link enhancement.

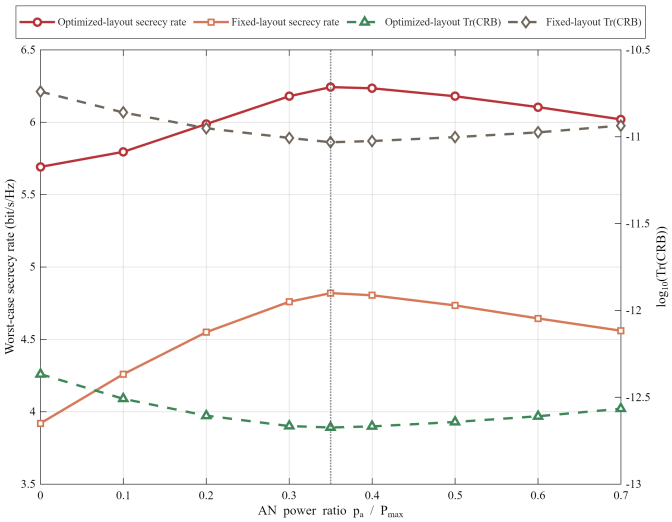


Fig. 3. Secrecy-sensing tradeoff versus the AN power ratio $p_a/P_{\max}$.

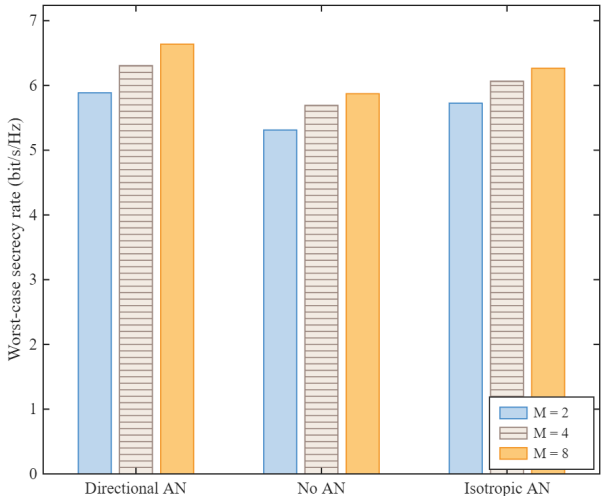


Fig. 4. Worst-case secrecy rate with different AN strategies under different transmit-antenna numbers.

B. Secrecy-Sensing Tradeoff Under AN Power Control

Fig. 3 shows that FA reconfiguration is not a marginal adjustment, but a way to enlarge the secrecy-sensing operating region. As $p_a/P_{\max}$ increases, the secrecy rate first improves and then saturates, while the CRB is most favorable in a moderate AN-power region. The marker at $p_a/P_{\max} \approx 0.35$ therefore indicates a balanced operating point rather than a maximum-power design. More importantly, the optimized-layout scheme consistently achieves both a higher secrecy rate and a smaller CRB than the fixed-layout baseline across the entire sweep, which means that geometry reconfiguration simultaneously improves secrecy and sensing by reshaping the near-field propagation structure.

C. AN Strategy Comparison

Fig. 4 further compares the effects of AN strategy and transmit-antenna number on secrecy performance. Directional AN achieves the highest secrecy rate in all tested transmit-side configurations, isotropic AN remains second, and the no-AN baseline performs the worst, confirming that properly shaped AN suppresses unfavorable leakage directions more effectively than isotropic radiation or no AN injection. In addition, the secrecy rate increases with the transmit antenna number M for all three strategies, because a larger transmit array provides more spatial degrees of freedom for both legitimate-link enhancement and directional leakage suppression. Therefore, directional AN and larger transmit arrays provide an additional secrecy gain once a favorable geometry has been obtained.

V. Conclusion

This paper investigated a fluid-antenna-aided near-field secure ISAC system under eavesdropper location uncertainty. By jointly optimizing transmit and receive FA positions, communication beamforming, and directional AN under CRB, power, region, and spacing constraints, we formulated and solved a geometry-aware secrecy-sensing design problem via a mixed PSOSC-based algorithm. Numerical results showed that uncertainty lowers the secrecy ceiling, FA reconfiguration enlarges the secrecy-sensing operating region over fixed layouts, and directional AN together with larger transmit arrays further strengthens this gain. These findings suggest that, in near-field secure ISAC, array geometry should not be treated as a fixed hardware condition. Instead, under eavesdropper uncertainty and sensing requirements, it becomes an active design resource for secrecy-sensing integration.

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